1. Use the following Monte Carlo estimator to approximate the expected value $I = E[\exp(\sqrt{U})]$ where, $U \sim U(0,1)$:

$$I_M = \frac{1}{M}\sum_{i=1}^{M} Y_i, \text{ where } Y_i = \exp(\sqrt{U_i}) \text{ with } U_i \sim U(0,1).$$

   Take the values of M to be $10^2, 10^3, 10^4$ and 10^5. Determine the 95% confidence interval for I for all the four values of M that you have taken. What is the exact value of I? Compare the exact value of I with the estimated values of I for different values of M.

2. Consider the expectation $I = E[\exp(\sqrt{U})]$ where, $U \sim U(0,1)$. Use the following antithetic method to approximate I:

$$\widehat{I}_M = \frac{1}{M}\sum_{i=1}^{M} \widehat{Y}_i, \text{ where } \widehat{Y}_i = \frac{\exp(\sqrt{U_i}) + \exp(\sqrt{1-U_i})}{2} \text{ with } U_i \sim U(0,1).$$

   Take the values of M to be $\frac{10^2}{2}, 10^2, \frac{10^3}{2}, 10^3, \frac{10^4}{2}, 10^4, \frac{10^5}{2}$ and 10^5. Determine the 95% confidence interval for $\widehat{I}_M$ for all the values of M that you have taken.

3. Present the results that you have obtained in Question 1 and Question 2 above in a tabular form. Your table must consist of the values of estimates (using the two methods), 95% confidence intervals for I (from two methods), and the ratio of widths of both the intervals. How do the estimates of I using the two methods compare with the actual value of I?

Submission Deadline: September 26, 2026, 11:59 PM